

Junior FDE Pre-screening Assignment

Design and Implementation of a Multi-Agent System

Public GitHub Link Submission: **Wednesday, May 20th at 11:59 PM ET**, via e-mail to tyler.parks@wipro.com

In-Person Presentation: **Thursday, May 21st at 1:30 PM ET**, virtual presentation

Objectives

The goal of this assignment is to **design and implement a live multi-agent system** that demonstrates agent collaboration, task decomposition, and safe use of AI/LLMs. You will focus on architecture, security guardrails, and implementation strategy, rather than only model output.

- You may use your choice of AI/LLMs, foundation models, autonomous agents, or collaborative agent frameworks.
- You may use any cloud provider (AWS, Azure, or GCP are preferred) to host your system, which should be **accessible via a live internet link** (public endpoint, hosted UI, or API).
- The project must be **presented virtually**, either:
 - Through a live demonstration using the deployed link, or
 - Via a recorded video walkthrough showcasing the system, its architecture, and functionality.

Problem Statement and Use Cases

Design a multi-agent system that collaboratively addresses a non-trivial use case. For example: wealth management, fraud analysis, information retrieval, daily planning, decision-making, customer support, data analysis, workflow automation, research and report generator, software/coding assistance, hiring assistance, or any comparable real-world system.

The system should consist of multiple specialized agents that interact with each other to achieve a shared goal. You are encouraged to be creative, but designs must be realistic and defensible. This open-ended assignment emphasizes “system thinking” over raw model output.

Requirements

Your submission must address the following areas:

1. Multi-Agent Architecture

Describe and justify your system design:

- Number and types of agents (e.g., Planner, Executor, Critic, Security Agent, User Interface Agent).
- Responsibilities and boundaries of each agent.
- Communication patterns (e.g., message passing, shared memory, orchestration layer).
- Whether agents operate sequentially, in parallel, or hierarchically.

2. Security, Safety, and Guardrails

Explain how your system addresses security and compliance, including:

- Input validation and prompt injection protection.
- Guardrails for LLM usage (role constraints, output filtering, policy enforcement).
- Data handling considerations (PII, secrets, logging).
- Measures to prevent unintended agent actions or escalation.

3. Implementation Approach

Detail how you planned and built the system:

- Tools, frameworks, or libraries used (e.g., Python, LangChain/LangGraph, AutoGen, CrewAI, Semantic Kernel, custom orchestration).
- How agents are instantiated, coordinated, and terminated.
- How the system handles errors, retries, and failures.
- Any evaluation or testing approach used to verify correctness.

4. Use of AI / LLMs and Collaboration

Explain:

- How and where LLMs are used (e.g., reasoning, planning, summarization, critique).
- How agents collaborate or negotiate decisions.
- Trade-offs between autonomy vs. control in your design.

Deliverables

Required:

- In-person presentation of the project.
- Written report (1–2 pages max) covering the sections above.
- Link to a public repository containing any project material (recorded video, write-ups, etc.).
 - ***Reply to the e-mail containing this assignment.***

Optional:

- Architecture diagram.
- Sample prompts.
- A recorded video walkthrough.

Evaluation Criteria

You and your project will be evaluated on:

- Clarity and soundness of the multi-agent design, including the responsible use of AI/LLMs.
- Thoughtfulness and practicality of security, guardrails, and implementation approach.
- Overall coherence and technical depth.